

Development of partial life-cycle experiments to assess the effects of endocrine disruptors on the freshwater gastropod *Lymnaea stagnalis*: a case-study with vinclozolin

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Abstract Long-term effects of endocrine disruptors (EDs) on aquatic invertebrates remain difficult to assess, mainly due to the lack of appropriate sensitive toxicity test methods and relevant data analysis procedures. This study aimed at identifying windows of sensitivity to EDs along the life-cycle of the freshwater snail *Lymnaea stagnalis*, a candidate species for the development of forthcoming test guidelines. Juveniles, sub-adults, young adults and adults were exposed for 21 days to the fungicide vinclozolin (VZ). Survival, growth, onset of reproduction, fertility and fecundity were monitored weekly. Data were analyzed using standard statistical analysis procedures and mixed-effect models. No deleterious effect on survival and growth

occurred in snails exposed to VZ at environmentally relevant concentrations. A significant impairment of the male function occurred in young adults, leading to infertility at concentrations exceeding 0.025 µg/L. Furthermore, fecundity was impaired in adults exposed to concentrations exceeding 25 µg/L. Biological responses depended on VZ concentration, exposure duration and on their interaction, leading to complex response patterns. The use of a standard statistical approach to analyze those data led to underestimation of VZ effects on reproduction, whereas effects could reliably be analyzed by mixed-effect models. *L. stagnalis* may be among the most sensitive invertebrate species to VZ, a 21-day reproduction test allowing the detection of deleterious effects at environmentally relevant concentrations of the fungicide. These results thus reinforce the relevance of *L. stagnalis* as a good candidate species for the development of guidelines devoted to the risk assessment of EDs.

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Introduction

Endocrine disruptors (EDs) have been distinguished from other toxic compounds because of their specific mode of action and effects on reproduction and development (Colborn et al. 1993). EDs interfere with synthesis, secretion, transport, association, action or elimination of natural hormones (Segner et al. 2003), inducing inappropriate responses of hormone targets. They are known to generate a variety of alterations in the development and functioning of the reproductive system in human and animal

populations (Toppari et al. 1996; Gist 1998; Sharpe and Irvine 2004), such as feminization of males, delayed development of male sexual organs, less mobile spermatozooids, etc., as reviewed in Matthiessen (2003). Precocious female development, alteration of vitellogenin synthesis and/or use, or imposition of male characteristics in females (in molluscs) are other consequences of exposure to EDs, which may lead to changes in reproductive success (Matthiessen 2003).

Sublethal effects of EDs on development and reproduction may be difficult to assess using current toxicity test methods because those effects may be induced by low doses of compounds (NIEHS 2001), and generally appear over the long-term (Bryan et al. 1987, 1991), so that acute toxicity tests may not detect them. Furthermore, biological responses may not exhibit classical dose-dependence relationships, but more complex patterns (e.g. inverted U dose-response curves) including both quantitative and qualitative changes depending on exposure concentration, as shown e.g. for the estrogenic compound bisphenol A (Weltje et al. 2005) and for the anti-androgen vinclozolin (Gray et al. 1994). Because of these particular properties of EDs, current toxicity test protocols and standard data analysis methods may lead to unrealistic risk forecasts for these compounds (Chen 2001; Matthiessen and Johnson 2007). Indeed, toxicity tests often focus on the assessment of short-term mortality and/or development of early life stages of model species which are supposed to be the most sensitive to a wide range of toxicants (OECD 1998). Therefore, such toxicity tests may not expose organisms during the phases of their life-cycle which are most responsive to EDs, nor measure endpoints that are sensitive to EDs. Moreover, the inability for high-dose tests to predict low-dose results has been pointed out by several publications (e.g. Myers et al. 2009). Consequently, the Organisation for Economic Cooperation and Development (OECD) has promoted the development and validation of new guidelines for EDs, in line with its Conceptual Framework for the Testing and Assessment of Endocrine Disrupting Chemicals (Gourmelon and Ahtiainen 2007). By the end of 2010, validated guidelines ranging from screening procedures to partial life-cycle experiments should be produced for rodents, amphibians, fish and invertebrates. This battery should be sensitive to a wide range of compounds, including inter alia (anti)estrogens, (anti)androgens, aromatase inhibitors, juvenile hormone mimics, and ecdysone agonists and antagonists.

Among invertebrates, molluscs (and more particularly gastropods) were identified as promising model species for the development of such guidelines, next to insects (e.g., *Chironomus riparius*) and crustaceans (e.g., *Daphnia magna*) (DeFur et al. 1999; Oehlmann et al. 2007; Matthiessen 2008). The fresh/brackish-water mesogastropod *Potamopyrgus antipodarum* and the freshwater gastropod

Lymnaea stagnalis can be considered as promising candidates to develop toxicity test procedures, which could eventually form the basis of forthcoming international guidelines (Matthiessen 2008). Furthermore, *L. stagnalis* currently appears as a good candidate to investigate possible modes of action of EDs because the endocrine system has been more largely investigated in this species than in other snails (Lagadic et al. 2007). However, there is a need for optimization and validation of toxicity test procedures with this snail before it can be routinely used in higher-tier protocols for the ecological risk assessment of EDs. Indeed, literature about the effects of EDs in *L. stagnalis* is still scarce (see e.g., Lagadic et al. 2007 and Matthiessen 2008). Moreover, previous work sometimes led to conflicting results regarding reproductive effects of the compounds, as for (i) the estrogen ethinyl-estradiol (EE-2), where the results from Segner et al. (2003) and Weltje et al. (2003) are not in full accordance, and (ii) for the androgen methyltestosterone, where the results from Czech et al. (2001) differ from those of Casey et al. (2005). These discrepancies are probably due to variations in methodologies such as starting the experiments with either adults or egg-clutches (Matthiessen 2008).

In this context, our study aimed at identifying windows of sensitivity to EDs along the life-cycle of *L. stagnalis*, and corresponding sensitive endpoints. Therefore, partial life-cycle (PLC) experiments were proposed to assess endocrine disrupting effects of vinclozolin in various development stages of the snail: juveniles (immature snails), sub-adults (snails that were close to sexual maturity but not fully mature at the beginning of the PLC experiment), young adults (snails that reached maturity and began to reproduce during the experiment), and adults (snails which had regularly reproduced before the beginning of the PLC experiments). Survival, growth, development and reproduction were monitored during 21 days of exposure for each development stage, in order to determine accurately sensitive endpoints for the studied development stage.

The anti-androgen vinclozolin (VZ; 3-(3,5-dichlorophenyl)-5-methyl-5-vinyl-1, 3-oxazolidine-2,4-dione; CAS number 50471-44-8) was chosen as a representative of its chemical family, because it has been shown to reduce reproductive success in various species of gastropods (Tillmann et al. 2001). VZ and its M1 and M2 metabolites are potent in vitro and in vivo anti-androgenic compounds in mammals, inhibiting androgen receptor binding and gene expression (Wong et al. 1995; Kelce and Wilson 1997; Kojima et al. 2004). M2 appears to be more potent than both the M1 metabolite and the parent compound (Kelce et al. 1994). VZ has also been shown to induce cytochrome P450-dependent aromatase activity (CYP19) which converts testosterone into estradiol (Sanderson et al. 2002) and which has been identified in several molluscs

species (Sternberg et al. 2010). Therefore, due to the reprotoxic properties of VZ and its metabolites, all commercial products containing this active ingredient have been banned in Europe since 2006. Nevertheless, VZ is still widely used as a fungicide in other countries, including the USA, to treat grapes, fruits, vegetables, turf grass, and ornamental plants against diseases caused by *Botrytis cinerea*, *Sclerotinia sclerotiorum*, and *Monilinia* spp. (US-EPA 2000). Chronic risk of VZ to aquatic organisms has not been assessed yet, due to a lack of data (US-EPA 2000). In our study, snails were thus chronically exposed to a large range of VZ concentrations (from 0 to 2,500 µg/L).

Data were analyzed using two different statistical methods: (i) standard analysis procedures as described in OECD guidelines (OECD 2006), and (ii) mixed-effect models. These latter models were implemented in order to provide comprehensive information on VZ effect patterns, especially with respect to the interaction between concentration and exposure duration. The ecotoxicological relevance of the proposed PLC experiments, chosen endpoints, and data analysis methods are discussed in the context of ecological risk assessment of EDs.

Materials and methods

Experimental populations

Lymnaea stagnalis (Linnaeus, 1758) (Gastropoda, Pulmonata, Basommatophora) is an herbivorous freshwater snail that is common in the northern hemisphere and abundant in ponds, lakes and ditches (Berrie 1965). Its ecology has been widely described (e.g., Berrie 1965, Kolodziejczyk and Martynuska 1980), and its endocrinology (e.g. Vreugdenhil et al. 1985) and reproductive biology (e.g. Geraerts and Joosse 1984) have been the subject of extensive investigations. This simultaneous hermaphroditic snail mainly reproduces by crossed-fertilization, but selfing may also occur in particular conditions (Jarne et al. 1993). This snail is reared at the INRA Experimental Unit of Aquatic Ecology and Ecotoxicology (Rennes, France) under laboratory conditions as previously described (Coutellec and Lagadic 2006).

From September to December 2008, 50 egg-clutches were collected every month from the same genitor batch and settled in independent 30 L aquaria in order to obtain cohorts of snails at different development stages for the PLC experiments. Snails were grown during 5 months in dechlorinated, charcoal-filtered tap water under controlled temperature ($21 \pm 1^\circ\text{C}$) and photoperiod (14/10 L/D). They were fed ad libitum with organic lettuce. Snails from the various cohorts were sorted according to their maximal

shell length in order to select organisms of homogenous size at the beginning of the PLC experiments: juveniles (9.93 ± 0.3 mm), sub-adults (14.6 ± 0.4 mm), young adults (20.8 ± 0.8 mm) and adults (29.8 ± 0.2 mm). Snails were gently transferred to test chambers containing culture water, 24 h prior to the exposure to VZ.

Vinclozolin treatments

Due to the scarcity of data on long-term effects of VZ on aquatic organisms, snails were exposed to a wide range of VZ concentrations. Tested concentrations (0; 0.025; 0.25; 2.5; 25; 250 and 2,500 µg/L) were chosen based on (i) previous studies on snails (Tillmann et al. 2001) and fish (Kiparissis et al. 2003), and (ii) pre-experiments with *L. stagnalis* (Weltje et al. 2003; Lagadic et al. 2005). VZ was purified from the commercial preparation Ronilan® (BASF Corp., Agricultural products, Mount Olive NJ, USA). VZ crystals (97.6% purity, as estimated by HPLC measurements) were dissolved in dechlorinated, charcoal-filtered tap water in order to obtain the appropriate nominal concentrations. Acclimation water was contaminated by pouring the appropriate volume of VZ solution in each test chamber, using a micropipette, after having removed an equivalent volume of clean water.

As proposed in standard semi-static toxicity test methods (OECD 1998), exposure water was renewed weekly during the experiments in order to eliminate snail faeces and ensure appropriate physico-chemical characteristics regarding O_2 , NH_4^+ , NO_2^- and HPO_4^- contents. Because VZ exhibits a 1.3-day half-life in an aqueous solution of pH 7 at 20°C (FOOTPRINT Pesticide Properties DataBase; FOOTPRINT 2006), it was expected to dissipate between water renewals, producing M1 and M2 metabolites at rates which may differ from one renewal cycle to the other. To avoid a sudden and unrealistic change in chemical composition and concentration of exposure water when renewing it, renewal water was prepared at the same time as treatment solutions, and stored until use in tanks that contained neither snails, nor food. VZ dissipation rate was assumed similar in exposure and renewal water.

Partial life-cycle experiment design and measured life-cycle traits

Groups of five snails were exposed to VZ in 0.5 and 1 L test chambers for juveniles and adults, respectively. Depending on the number of snails of appropriate length in each cohort at the beginning of the experiments, three, four and six replicates were implemented for sub-adults, young adults, and juveniles and adults, respectively, for each tested concentration. Snails were fed ad libitum with

calibrated slices of organic lettuce. Food quantities were defined based on previous experiments (Ducrot et al. 2010) and corresponded to 0.160 (± 0.03), 0.475 (± 0.075), 0.7 (± 0.1) and 2.0 (± 0.1) g per group of five snails per day, for juveniles, sub-adults, young adults and adults, respectively.

Effects of VZ on survival, growth and reproduction were monitored at each development stage during 21 days (i.e., maximal duration between two successive development stages in our experimental conditions), so that the PLC experiments covered 25% of snail life-span. Dead snails were counted weekly and withdrawn from the test chambers. Growth was estimated from the mean shell length values, measured using a digital calliper. Effects on reproduction were estimated based on both fertility (ability to produce viable eggs) and cumulated fecundity (cumulated number of eggs produced per individual). Fecundity was measured daily by collecting and counting egg-clutches in each test chamber. Each clutch was then transferred to a Petri dish and settled in 10 mL of culture water. The number of eggs per clutch was determined using a stereomicroscope.

Data analysis

Experimental data were first analyzed using standard nonparametric statistical analysis procedures, as described in OECD (2006). Significant effects of VZ concentration on final survival, and on mean shell length and mean cumulated fecundity at days 7, 14 and 21 were first highlighted using Kruskal–Wallis tests. No Observed Effect Concentration (NOEC) and Lowest Observed Effect Concentration (LOEC) were estimated using Mann–Whitney tests with a Bonferroni adjustment. The effect of exposure duration was tested using a Friedman test.

Then, the data were analyzed as functions of both VZ nominal concentration and exposure duration, in order to assess effects of VZ on the response patterns for each studied endpoint. Therefore, the survival pattern in exposed snails was assessed using Kaplan–Meier survival curves. A log-rank test was also performed to determine the effect of VZ concentrations on snail survival throughout the experiment. Effects on growth and fecundity were analyzed using mixed-effect models, depending on two fixed factors (exposure duration and VZ concentration) and one random factor (replicate). Growth and reproduction variables were non-linear according to exposure duration, which was thus considered as a qualitative factor with three modalities (7, 14 and 21 days). Effects of factors on survival, growth and reproduction were considered statistically significant at $p < 0.05$. Statistical analyses were performed with the “survival” and “nlme” packages of the software R (R Development Core Team 2007).

Results

Effects of VZ on snail survival

No juvenile or sub-adult died during the experiment, showing that VZ did not hamper survival in immature organisms. Mortality occurred in young adults (for 5 and 15% of the snails exposed during 21 days at 2.5 and 2,500 $\mu\text{g/L}$, respectively), but final mortality rates were not significantly different from the control value. However, significant mortality occurred in adults (Kruskal–Wallis test, $p = 0.017$ at day 21) exposed to 2,500 $\mu\text{g/L}$ (Mann–Whitney test with Bonferroni adjustment, $p = 0.006$). Theoretical survival probabilities over 21 days were represented using Kaplan–Meier survival curves (Fig. 1). The log-rank test showed a significant effect of VZ on survival pattern ($p < 0.01$), probably due to the high mortality rates observed at the highest tested concentration.

Effects of VZ on snail feeding activity and growth

Snails exposed to 250 and 2,500 $\mu\text{g/L}$ exhibited behavioural troubles, gathering on the bottom of test chambers, and decreasing their feeding activity (i.e. searching and consumption of food). Snails exposed to 2,500 $\mu\text{g/L}$ even stopped feeding. The analysis of shell length data at days 7, 14 and 21 with the standard approach showed no significant effect of VZ on exposed snails. Growth patterns in control snails were in adequation with the theoretical growth in

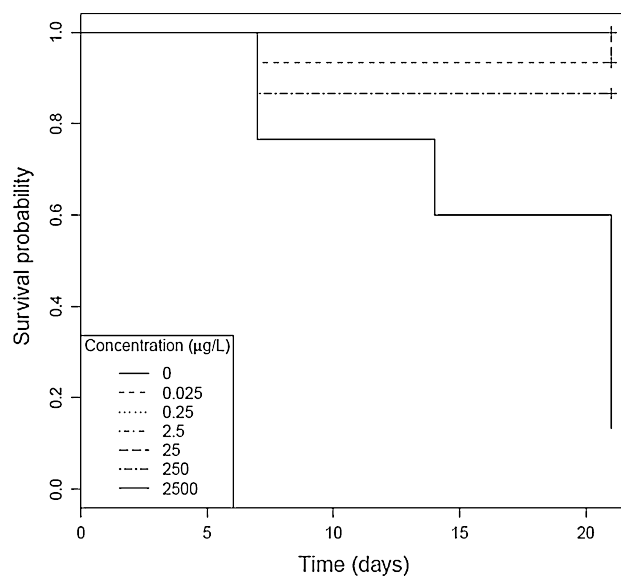


Fig. 1 Survival probabilities, as described by Kaplan–Meier survival curves, in adults of *Lymnaea stagnalis* exposed to vinclozolin during 21 days in a partial life-cycle experiment. The plus sign (+) at the end of curves represents right censored data

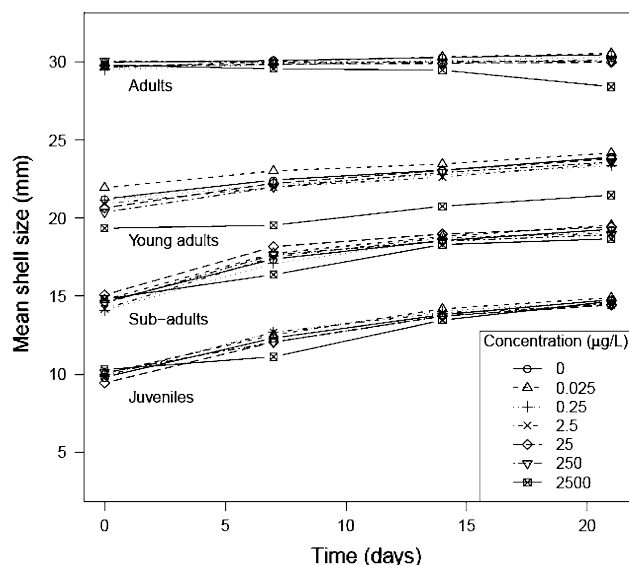


Fig. 2 Evolution of the mean shell size in successive development stages of *Lymnaea stagnalis*, after 21 days of exposure to vinclozolin in partial life-cycle experiments. Errors bars were removed in order to facilitate reading

L. stagnalis (Von Bertalanffy growth, as described by Zonneveld 1992). However, growth patterns in exposed snails were modified according to their development stage (Fig. 2). For example, all snails (except adults) exposed to 2,500 µg/L VZ grew slower than the controls during the first week of exposure, but partly recovered during the second and third weeks, so that the transient deviation from the control growth pattern did not lead to a significant decrease in the mean shell size after 21 days of exposure. Complementary analysis of shell length data with mixed-effect models highlighted a significant interaction between exposure duration and VZ concentration for each development stage (p -values were all below 0.01). This result demonstrated the occurrence of significant effects of VZ on the growth pattern of exposed snails, and statistically confirmed what was previously interpreted from Fig. 2. Such transient effects on growth were not detected when applying the standard statistical analysis to the data.

Effects of VZ on snail reproduction

Juveniles and sub-adults did not reproduce during the experiments, because they did not reach sexual maturity (as indirectly defined by reaching the mean threshold size at first spawning in our experimental conditions: 22.2 ± 2 mm, V. Ducrot, unpublished results). Within young adults, spawning occurred from day 13 to the end of the experiment, in all snails except those exposed to 2,500 µg/L, which did not reproduce. Due to their youth, they exhibited a low and highly variable fecundity: only 31.8 ± 24.0 eggs were laid in average per control snail per

week. Mean cumulated fecundity of controls was reduced by 31, 54, 77 and 100% in young adults exposed to nominal VZ concentrations of 0.025, 0.25, 25 and 2,500 µg/L, respectively, showing that cumulated fecundity was significantly reduced even in snails exposed to low doses of VZ (Kruskal–Wallis test, $p < 0.01$). Furthermore, the eggs contained in those clutches were generally not fertilized. Non-fertilized eggs (diffuse and whitish nucleus, no movement) were easily distinguished from developing embryos (compact yellow embryos, slow movements). This highlighted a pronounced impairment of the male function, leading to infertility in VZ-exposed young adults.

Fertility was not impaired in adults, but VZ induced a decrease in cumulated fecundity in exposed snails over 21 days (Kruskal–Wallis test, $p = 0.01$). This decrease was significant only in snails exposed to concentrations exceeding 250 µg/L (LOEC value, as determined by Mann–Whitney tests with Bonferroni adjustment, $p = 0.002$). The cumulated number of eggs produced per snail per week remained constant over the experiment in control snails and in snails exposed to low VZ concentrations, whereas effects of VZ on fecundity were the most pronounced during the first week of exposure in snails exposed to concentrations exceeding 25 µg/L (Fig. 3). However, the effect of exposure duration was only significant in snails exposed to 2,500 µg/L (Friedman test, $p = 0.04$). Recovery occurred during the experiment among snails exposed to 25 µg/L, which exhibited a cumulated fecundity equivalent to that of the controls at the end of the test, but did not occur in snails exposed to higher VZ concentrations.

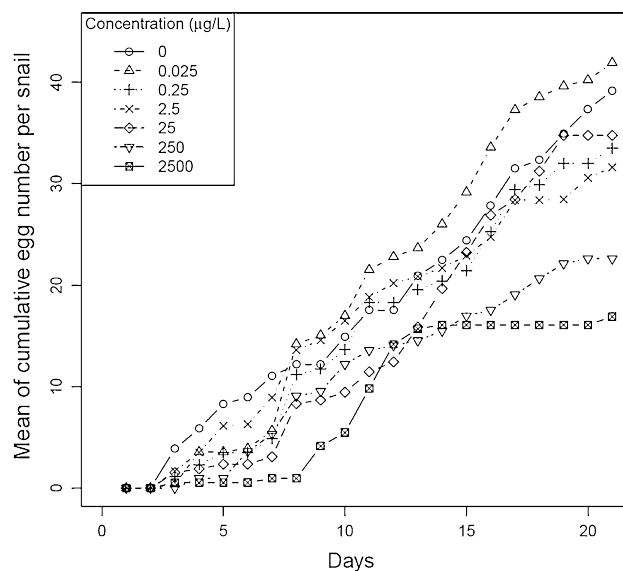


Fig. 3 Mean cumulated fecundity in adults of *Lymnaea stagnalis* exposed to vinclozolin during 21 days in a partial life-cycle experiment. Errors bars were removed in order to facilitate reading

Based on these results, the concentration–response relationship was complex. Indeed, different biological responses were observed between the snails exposed to low ($<25\text{ }\mu\text{g/L}$) and high concentrations ($25\text{--}2,500\text{ }\mu\text{g/L}$) of VZ. Differences in recovery potential occurred in this latter group of snails. Furthermore, observed effects varied with exposure duration for a given concentration. As confirmed by the analysis of cumulated fecundities using mixed-effect models, snail fecundity pattern significantly depended on both exposure duration and VZ concentration ($p < 0.01$). Effects on fecundity occurred when VZ concentration reached $25\text{ }\mu\text{g/L}$, although this concentration was supposed to be harmless for the snails based upon the results of standard statistical analysis of the reproduction data at 21 days ($\text{LOEC} = 250\text{ }\mu\text{g/L}$).

Discussion

Sensitivity and biological responses of *L. stagnalis* to VZ

Based on the results of PLC experiments, *L. stagnalis* is more sensitive to VZ than the standard model animal species for ecological risk assessment of aquatic ecosystems. For example, the lowest reported LOEC values are $60\text{ }\mu\text{g/L}$ in 21-day fecundity experiments with adult fat-head minnows (Martinović et al. 2008), and $1,000\text{ }\mu\text{g/L}$ in 21-day reproduction experiments with *Daphnia magna* (Haeba et al. 2008). No data are currently available for Chironomidae. Our study thus confirms that gastropods appear to be among the most sensitive aquatic animal species to VZ, as suggested by Matthiessen (2008). Indeed, VZ was found to induce significant effects on the reproduction of young adults of *L. stagnalis* at environmentally relevant concentrations ($\text{LOEC} = 0.025\text{ }\mu\text{g/L}$, whereas maximal VZ concentration detected in surface water is $0.5\text{ }\mu\text{g/L}$, Glden et al. 1997 in Tillmann et al. 2001). Therefore, *L. stagnalis* may be sensitive enough to detect effects of EDs at realistic environmental concentrations.

Our study suggests that *L. stagnalis* may be among the most sensitive gastropod species to VZ exposure. Indeed, Tillmann et al. (2001) studied the effects of VZ on two other snails, which were known to be sensitive to various EDs. In their study, sub-adults of the freshwater gastropod *Marisa cornuarietis* and the marine prosobranch *Nucella lapillus* were exposed for 5 months to VZ concentrations ranging from 0.03 to $1.0\text{ }\mu\text{g/L}$. No mortality was reported, but a significant decrease in the length of the penis and accessory male sex organs was observed in both species. These effects occurred during the first month of exposure to the lowest tested VZ concentrations (0.03 to $0.1\text{ }\mu\text{g/L}$), and were reversible. Male fertility was impaired at a

similar concentration range in *L. stagnalis* and in these two other species, thus suggesting that reproductive responses of the pond snail are in line with the responses of other sensitive snails. As in *M. cornuarietis*, the biological response patterns of *L. stagnalis* to VZ were complex. Indeed, growth and fecundity rates in snails exposed to a given concentration of VZ varied according to the exposure duration. Moreover, those rates did not monotonically decrease as a function of exposure concentration. This can be explained by the fact that effects are due to the simultaneous occurrence of VZ and its M1 and M2 metabolites which do not show production and elimination kinetics that are directly linked with VZ concentration (Kelce et al. 1994; Mercadier et al. 1998). Previous experiments showed that, after 48 h, M1 concentration in exposure water was three times higher than VZ concentration, whereas M2 concentration was four times lower. In *L. stagnalis*, after 96 h of exposure, VZ internal concentration represented $67.52 \pm 8.81\%$ of the total internal residue concentration, M1 and M2 internal concentrations being only $2.02 \pm 0.92\%$ and $6.79 \pm 2.20\%$, respectively (J.-P. Cravedi, unpublished results). However, the fact that VZ is the major residue in snail tissues does not mean that it is responsible for the observed effects since M1 and M2 metabolites are respectively 10 and 100 times more potent than VZ with respect to anti-androgenicity (Kelce et al. 1994). In addition, the reprotoxicity of three polar residues also found in snail tissues (ca. 23.5% of the total internal residue concentration after 96 h of exposure; J.-P. Cravedi, unpublished results) remained unknown. To a lesser extent, growth and reproduction patterns might have been transiently modified, via a decrease in feeding rate in snails exposed to VZ concentrations exceeding $250\text{ }\mu\text{g/L}$, which exhibited abnormal feeding behaviours. Such behavioural changes, that affect locomotor activity and food consumption, have been described in rats exposed to dietary VZ (Flynn et al. 2001).

Life-cycle tests with *L. stagnalis* exposed during 130 days to $0.286\text{ }\mu\text{g/L}$ of VZ from the juvenile to the adult stage (Weltje et al. 2003) showed that sexual maturity occurred at 19.5 mm in both exposed and control snails, which is confirmed by our study (i.e. size at sexual maturity was not modified in VZ-exposed snails). At $0.286\text{ }\mu\text{g/L}$, VZ significantly stimulated fecundity in exposed snails as compared to controls (Weltje et al. 2003), which was observed at $0.025\text{ }\mu\text{g/L}$ in our study (see Fig. 3). Such an increase in fecundity was assumed to be due to a repression of the male function due to the anti-androgenic mode of action of VZ (Weltje et al. 2003). Yet, such a stimulation of egg production was not observed at higher concentrations in our study, as confirmed by previous studies on adults we conducted under similar exposure conditions (Lagadic et al. 2005). These results

reinforce the hypothesis that VZ probably induces changes in the male function of exposed snails, as demonstrated in fish (e.g., Baatrup and Junge 2001; Bayley et al. 2003; Kiparissis et al. 2003) and rat (e.g., Gray et al. 1999; Kubota et al. 2003), the observed changes depending on exposure concentration. Full life-cycle (FLC) and PLC experiments did not lead to similar conclusions for close values of VZ concentration (i.e. 0.286 and 0.250 $\mu\text{g/L}$ in the FLC and PLC experiments, respectively). This result confirms that the test design might greatly influence test outputs (Matthiessen 2008).

Optimizing sensitivity of PLC experiments with *L. stagnalis* for risk assessment of EDs

Significant mortality only occurred in adults exposed to the highest tested concentration (2,500 $\mu\text{g/L}$). This result confirmed previous studies on various development stages of the fathead minnow (Makynen et al. 2000; Martinović et al. 2008), and in adults of other gastropod species (Tillmann et al. 2001), which showed that VZ had no effect on survival at environmental concentrations. Additionally, in the present study, differences observed between the growth pattern of control snails and those exposed to environmentally relevant VZ concentrations were not found to be statistically significant after 21 days of exposure. This suggests that survival and growth may not be appropriately sensitive endpoints to detect effects of anti-androgens in *L. stagnalis*.

Young adults were particularly sensitive to VZ, because they exhibited both a significant reduction in fecundity (30% less than controls) and a significant decrease in the fertilization rate of these eggs at the lowest tested concentration (0.025 $\mu\text{g/L}$). The number of empty eggshells in egg-clutches that were laid by exposed snails was similar to the control value, indicating that the female function was not impaired by VZ. Fertilization failure was most likely due to the impairment of the male function which might have altered spermatogenesis in young adults due to the anti-androgenic action of VZ. Unlike young adults, adults exhibited no trouble in fertility. This is probably because *L. stagnalis*, like other pulmonate snails, can store and nourish allosperm for months after copulation (Fretter and Graham 1962): adults may have stored healthy sperm from previous copulation with snails from the culture, and may have used it to fertilize eggs during the PLC test. Adults were also less sensitive to VZ than young adults, since their fecundity was less impacted at similar exposure concentrations. Moreover, fecundity rate of adult snails could recover during the experiments if VZ concentration did not exceed 25 $\mu\text{g/L}$. Therefore, it is concluded that the age and maturation status of exposed snails significantly influenced the sensitivity of PLC experiments in which reproduction is

the endpoint, maximal sensitivity to anti-androgens being obtained by starting such experiments with young adults. These results showed that, for *L. stagnalis*, exposure to toxicants should start with snails that are not fully mature, so that PLC experiments could detect potential effects of EDs on spermatogenesis.

Based on these results, 21-day experiments with young adults of *L. stagnalis*, allowing measuring both fertility and fecundity, may constitute a sensitive approach to assess effects of anti-androgenic compounds in this species. Experiments should be implemented under standard laboratory conditions of temperature and light, and ad libitum feeding with organic lettuce (or appropriate commercial fish food if it does not contain EDs). Individual performances should be monitored weekly to ensure detecting transient effects of EDs (e.g., in our study, a decrease in fecundity at 25 $\mu\text{g/L}$ in adults can be detected at days 7 and 14, but not anymore at day 21), and recovery. In the case of fecundity impairment due to ED exposure, it might be necessary to extend the experiment duration in order to obtain enough data to avoid a loss of power in subsequent statistical tests. Furthermore, because EDs may have epigenetic effects (see Nilsson et al. 2008 for an example with VZ), complementary assessment of hatching success in the offspring of exposed snails is recommended. However, such experiments generate complex data sets, which require to be analyzed with appropriate statistical methods.

Statistical analysis of data from PLC experiments with EDs in *L. stagnalis*

In addition to the standard data analysis tools, other statistical tests and models were used to account for exposure duration in the analysis of VZ effects in *L. stagnalis*. All tests showed that this latter factor significantly contributed to explain the response patterns of exposed snails. Survival data were adequately described for each development stage using Kaplan–Meier curves. Transient growth effects could not be detected in a standard growth experiment (shell length measurement at the end of the experiment only) with a standard statistical analysis procedure. Only the analysis of a suite of several successive length measurements by mixed-effects models allowed detecting these effects. Similarly, transient effects on reproduction could not be detected using standard experimental and data analysis approaches. Furthermore, the use of the standard statistical approach to analyze fecundity data led to underestimation of VZ effects on reproduction, because the time-course of effects was neglected. Analyses with mixed-effect models provided a thorough description of the response pattern in VZ-exposed snails because mixed-effect models consider all factors and their potential interactions in the statistical analysis. Complex biological patterns, including both

quantitative and qualitative changes depending on exposure concentration, are adequately described using such a method. However, in the case of significant interaction between toxicant concentration and exposure duration, mathematical models which enable describing both toxicokinetics and toxicodynamics (TK/TD models) would be necessary to estimate no effect concentration values that do not depend on exposure duration (Hommen et al. 2010). Only this latter approach would ensure proper estimation of long-term effects of EDs and the corresponding effect thresholds. The development and use of mechanistic TK-TD models devoted to the risk assessment of EDs should thus be promoted.

Conclusions

PLC experiments highlighted that *L. stagnalis* was sensitive to environmentally realistic concentrations of VZ, confirming that this species was a good candidate to study the effects of anti-androgenic compounds. Further studies are required to assess its sensitivity towards other classes of EDs. Biological responses of *L. stagnalis* to VZ did not exhibit classical dose-dependence relations, but more complex patterns which included both quantitative and qualitative changes in the response, depending on exposure concentration, transient effects, and recovery potential in some concentration ranges. Such complex response patterns could not be captured using existing standard experiments. Therefore, we proposed a reproduction experiment, which consists in measuring fertility and fecundity in young adults, and which could be used as a relevant basis to optimize PLC experiment procedures with *L. stagnalis*, in order to eventually derive international guidelines devoted to risk assessment of EDs. Furthermore, the use of standard statistical methods to analyze growth and reproduction data obtained in PLC experiments led to the underestimation of the corresponding NOECs in sensitive life-stages of the snail. Mixed-effect models provided a comprehensive analysis of both the effect patterns and the relationships between effect factors, thus allowing further understanding of long-term responses in exposed snails. Moreover, mechanistic effect models (e.g., TK/TD models) devoted to EDs should be developed and used to estimate toxicity threshold values when concentration and exposure duration significantly interact in determining the biological response patterns of exposed organisms.

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